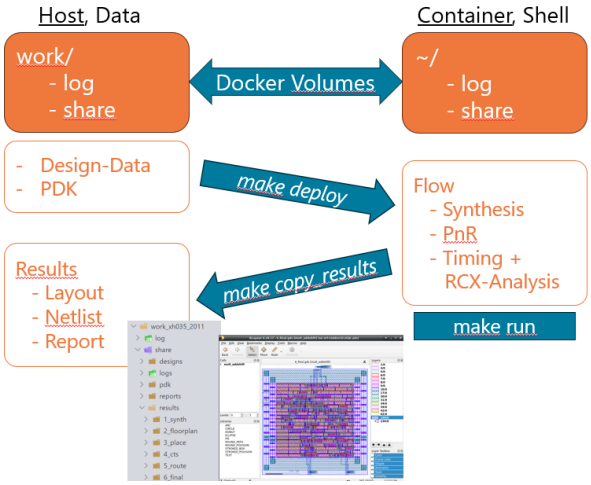
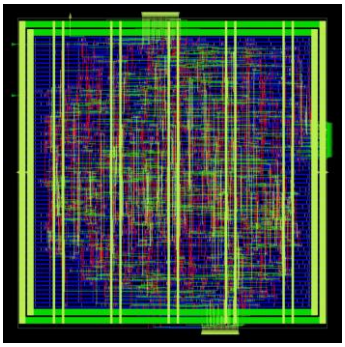
Developing an open source digital design flow using OpenROAD with X-FAB 350 nm technologies

To strengthen supply chains and make chip design available in a more liberal software environment, open source EDA tools are a great opportunity. Independent open source software tools especially benefit smaller businesses and individuals considering the rising costs for commercial software.

As associate in the funded project DI-Meta-X, X-FAB is developing an open source reference kit for users to showcase the use of open source EDA software for a complete digital chip design. OpenROAD is used as main toolchain software, which enables a complete RTL-to-GDS flow. Already existing flow structures from OpenROAD are adapted to fit X-FABs technology data and enable a fully automated digital design flow. This is demonstrated by the inclusion of XH035 as main technology for this reference kit. The complete kit is designed as containerized solution for rapid deployment in many environments.



Design flow inside the containerized reference kit



Example design given in the reference kit

With the reference kit the user is handed a completely configured OpenROAD platform for X-FABs XH035 technology for digital Place&Route operations. The fully automated RTL-to-GDS chain accepts Verilog as HDL input for any design. This then generates the final GDSII layout without any additional user input (provided user has done design configuration). Within the given platform configuration each resulting design is expected to pass DRC and LVS check with no further modifications necessary.

An automated and nearly fully configured digital design flow enables users to rapidly develop and test digital design iterations. With the additional benefit of open source software this enables broader adoption of chip design, especially for SMEs and individuals.

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